Review Test Submission: HAA 05

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Course	Fall 2024 Switching and Routing (NET-126-2001)
Test	HAA 05
Started	11/7/24 8:11 AM
Submitted	11/7/24 8:46 AM
Status	Completed
Attempt Score	100 out of 100 points
Time Elapsed	34 minutes out of 1 hour
Results Displayed	Submitted Answers, Incorrectly Answered Questions

Question 1

7 out of 7 points

Given the following host address determine the requested information:

180.22.188.33/19

Determine: **Custom Address Map (CAM)**

Selected Answer: N.N.ssshhhhh.H

Question 2

7 out of 7 points

Determine: **Total Number of Subnets Possible**

Hint: Total Subnets = 2^S, where s is the number of subnet bits in the CAM.

Selected Answer: 8

Question 3

7 out of 7 points

Determine: **Total Number of Addresses (per Subnet)**

Hint: Total Addresses = 2^h, where h is the number of host bits in the CAM.

Selected Answer: 8192

Question 4 7 out of 7 points

Determine: **Total Number of Usable Addresses (per subnet)**

Hint: Total Usable = 2^h - 2, where h is the number of host bits in the CAM.

Selected Answer: 8190

Question 5 7 out of 7 points

Determine: **Custom Subnet Mask**

Hint 1: Refer to the CAM. Turn on the network bits (set to 1) and turn off the host bits (set to 0).

Hint 2: Remember that subnet bits are network bits.

Selected Answer: 255.255.224.0

Question 6 7 out of 7 points

Determine: **Wildcard Mask**

Hint 1: Refer to the CAM. Turn off the network bits (set to 0) and turn on the host bits (set to 1).

Hint 2: Remember that subnet bits are network bits.

Selected Answer: 0.0.31.255

Question 7 7 out of 7 points

Determine: **Interval**

Hint: The interval is the value of the rightmost network/subnet bit within the octet.

Selected Answer: 32

Question 8 7 out of 7 points

Determine: **Transition Octet**

Hint: The Transition Octet (TO) is the octet where the last (rightmost) network/subnet bit resides (see Address Map).

Record your answer as 1st, 2nd, 3rd, or 4th.

Selected Answer: 3rd

Question 9 7 out of 7 points

Determine: **GTOV**

Hint: GTOV is the Given Transition Octet Value.

- Look at the **G**iven host address.
- Identify the **T**ransition **O**ctet (TO).
- The **V**alue in the **T**ransition **O**ctet is the GTOV.
- Example 1: The GTOV in 10.25.50.222/12 is 25.
- Example 2: The GTOV in 10.25.50.222/19 is 50.
- Example 3: The GTOV in 10.25.50.222/28 is 222.

Selected Answer: 188

Question 10 9 out of 9 points

Determine: **NATOV**

Hint: The NATOV is the Network Address Transition Octet Value. You will use it to easily determine the Network Address.

- Use the formula:

  NATOV = TRUNC (GTOV / INT) * INT

- GTOV / INT = the GTOV divided by the Interval (INT).
- TRUNC means to truncate the result - discard/ignore the digits after the decimal point.

- Example: 10.**204**.50.222/11

  - CAM: N.ss**s**hhhhh.H.H

- INT, TO = 32, 2nd
- GTOV = 204
- NATOV = TRUNC (GTOV / INT) * INT
- = TRUNC (204 / 32) * 32
- = TRUNC (6.375) * 32
- = 6 * 32
- = 192

- NATOV = Network Address Transition Octet Value
- GTOV = Given Transition Octet Value
- INT = Interval
- TO = Transition Octet

Selected Answer: 160

Question 11

7 out of 7 points

Determine: **Network Address and Prefix Length**

Hint: Use the NATOV to easily determine the network address.

- Look at the given host address.
- Replace the GTOV with the NATOV.
- Turn off (set to 0) all remaining host bits.

- Example for 10.204.50.222/12

  - CAM: N.sssshhhh.H.H
  - GTOV = 204
  - NATOV = 192
  - Network Address: 10.192.0.0/12

Selected Answer: 180.22.160.0/19

Question 12

7 out of 7 points

Determine: **First Usable Address**

Hint: Add 1 to octet 4 of the Network Address.

Do not include the prefix.

Selected Answer: 180.22.160.1

Question 13

7 out of 7 points

Determine: **Last Usable Address**

Hint: Determine the Broadcast Address first, then subtract 1 from octet 4 of the broadcast.

Do not include the prefix.

Selected Answer: 180.22.191.254

Question 14

7 out of 7 points

Determine: **Broadcast Address**

Hint: Network Address + Wildcard Mask = Broadcast Address

Do not include the prefix.

Selected Answer: 180.22.191.255

Thursday, November 7, 2024 6:20:11 PM EST

← **OK**